KONCHADA SURYAKSHAR

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SUMMARY

Electronics and Communication Engineering student with strong interest in digital design, FPGA-based systems, and embedded architectures. Experienced in Verilog-based digital system development, MATLAB-based communication system modeling, and sensor-driven embedded system design. Familiar with RTL design concepts, processor fundamentals, and hardware–software co-design for scalable and efficient digital systems.

WORK EXPERIENCE

Coincent AI
AI/ML Intern *April 2024 – May 2024*

- Implemented and optimized computational algorithms using **Python within structured software pipelines**.
- Performed **data processing, debugging, and performance evaluation** to improve system efficiency.
- Strengthened **analytical problem solving and system-level reasoning**, focusing on computational efficiency and scalable workflows.

PROJECTS

- **OFDM-Based Digital Communication System on Zynq FPGA *(FPGA Communication Systems)***
- Designed and implemented an end-to-end OFDM transmitter–receiver system targeting FPGA-oriented architectures using hardware–software co-design principles.
- Developed and validated baseband communication blocks in MATLAB, including QPSK modulation, FFT/IFFT processing, cyclic prefix insertion/removal, and signal flow analysis.
- Evaluated communication system performance through Bit Error Rate (BER) analysis under AWGN channel conditions, while analyzing real-time communication constraints, transmission reliability, and signal integrity.

High Beam Assist System *(Embedded Systems)*

- Designed an embedded hardware system using ESP32 and LDR sensors for automatic vehicle headlight control.
- Implemented real-time sensor-based decision logic for high-beam and low-beam switching.
- Integrated power MOSFET drivers and relay switching circuits to ensure reliable hardware operation.
- Focused on sensor interfacing, embedded control logic, and system-level integration.

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Smart Water Infrastructure Management System *(Verilog Digital Design)*

- Designed a Verilog-based digital controller for automated water management in apartment infrastructure.
- Implemented a Finite State Machine (FSM) to control pump operation, zone distribution, and fault handling.
- Developed modules for dry-run detection, overload protection, and sensor fault detection.
- Verified functionality using self-checking testbench simulations in Xilinx Vivado.

EDUCATION

Jawaharlal Technological University Hyderabad
Integrated (B.Tech + M.Tech) in **Electronics & Communication Engineering** 2022-2027
CGPA : 8.13 / 10

ADDITIONAL

TECHNICAL SKILLS: Digital electronics, Verilog, RTL concepts, Microcontrollers, embedded system design, Signal Processing, Finite State Machines(FSM),OFDM fundamentals, FFT/IFFT, C (basic), Python, MATLAB, Simulink.

CERTIFICATIONS: Hardware Modeling using Verilog (NPTEL – IIT Kharagpur), AI for everyone(Deep Learning ,Coursera), Python for Data Science and AI Development(IBM, Coursera), Machine Learning Specialization(Deep Learning.AI)

AWARDS/ACTIVITIES: Organized E-Papyrus Event – Spoorthi National Symposium , Participated in Smart India Hackathon 2024&2025,Participated in MSME Idea Hackathon 4.0 (got selected to 2nd level),2nd Prize, E-Papyrus Event – Spoorthi National Symposium (Group Presentation).

SOFT SKILLS: Problem-Solving, Critical Thinking, Analytical Skills, Adaptability, Teamwork, Communication